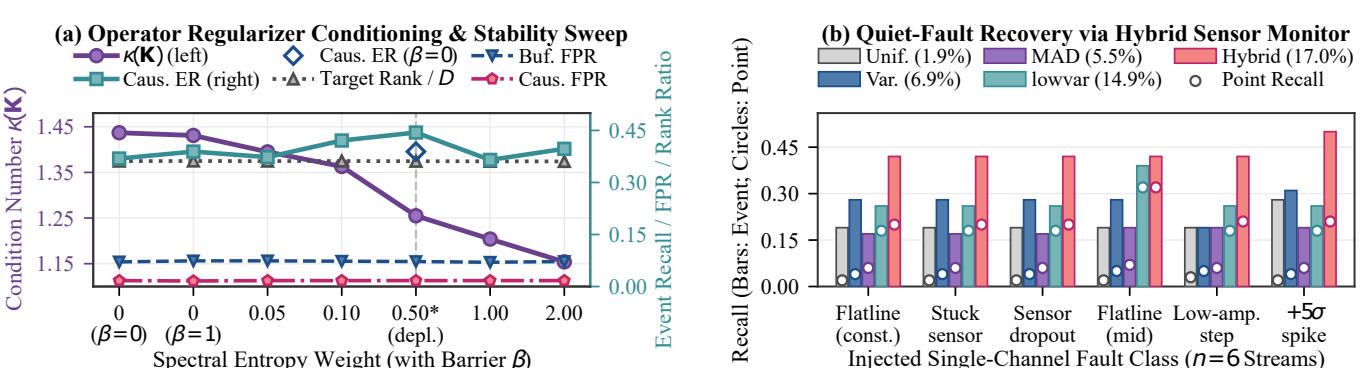

(a) Operator Regularizer Conditioning & Stability Sweep
κ(K) (left)
Caus. ER (β=0)
Buf. FPR
Caus. ER (right)
Target Rank / D
Caus. FPR
Condition Number κ(K)
1.45
1.35
1.25
1.15
0.45
0.30
0.15
0.00
Event Recall / FPR / Rank Ratio
0 (β=0)
0 (β=1)
0.05
0.10
0.50* (depl.)
1.00
2.00
Spectral Entropy Weight (with Barrier β)
(b) Quiet-Fault Recovery via Hybrid Sensor Monitor
Unif. (1.9%)
MAD (5.5%)
Hybrid (17.0%)
Var. (6.9%)
lowvar (14.9%)
Point Recall
Recall (Bars: Event; Circles: Point)
0.45
0.30
0.15
0.00
Flatline (const.)
Stuck sensor
Sensor dropout
Flatline (mid)
Low-amp. step
+5σ spike
Injected Single-Channel Fault Class (n=6 Streams)